

Question	Marking Guidance	Mark	Comments										
06.1	1. Small sample size; 2. Fusion/fertilisation of gametes is random; 3. Linked Genes; 4. Epistasis; 5. Lethal <u>genotypes</u> ;	2 max	2. Ignore breeding is random 3. Accept crossing over / sex linkage										
06.2	1. ttmm;	1	Accept mmtt or any order of these alleles e.g. mtmt, tmtm etc										
06.3	1. Genes are linked; 2. Produces few(er) tall, mottled and dwarf, normal offspring; 3. Crossing over (has occurred);	3	1. Accept 'Alleles are linked' but reject if context suggests alleles of the 'same gene' 2. Accept produces few Tm and tM gametes 2. Accept 'fewer recombinants'										
06.4	One mark for each correct column;; <table><tr><th>Phenotype of offspring</th><th>Ratio of offspring</th></tr><tr><td>Tall (plant and) normal (leaves)</td><td>9</td></tr><tr><td>Tall (plant and) mottled (leaves)</td><td>3</td></tr><tr><td>Dwarf (plant and) normal (leaves)</td><td>3</td></tr><tr><td>Dwarf (plant and) mottled (leaves)</td><td>1</td></tr></table>	Phenotype of offspring	Ratio of offspring	Tall (plant and) normal (leaves)	9	Tall (plant and) mottled (leaves)	3	Dwarf (plant and) normal (leaves)	3	Dwarf (plant and) mottled (leaves)	1	2	Note: Accept correct phenotypes in any order for one mark and correct ratio in any order e.g. 3:9:3:1 for one mark However, phenotypes and ratio must match for two marks Accept alternative wording e.g. short for dwarf 2. Accept $\frac{9}{16}$ $\frac{3}{16}$ $\frac{3}{16}$ $\frac{1}{16}$
Phenotype of offspring	Ratio of offspring												
Tall (plant and) normal (leaves)	9												
Tall (plant and) mottled (leaves)	3												
Dwarf (plant and) normal (leaves)	3												
Dwarf (plant and) mottled (leaves)	1												

